

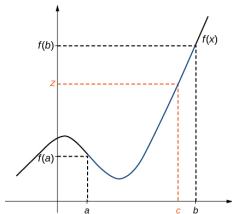
Week 03, Lecture 3

Continuity Types; The Intermediate Value Theorem

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These slides are by Niall Madden. Some content is based on notes by Dr Kirsten Pfeiffer. And some more, such as the figure opposite, taken from Strang & Herman's "Calculus". However, all the typos are Niall's.

Today we continue to learn about...

- 1 Continuity
- 2 Types of discontinuity
- 3 Intermediate Value Theorem
 - Examples
 - Application
- Some terminology
- Examples
- 4 Derivatives: the concept
 - Rate of change
- 5 Exercises

For more, see Chapter 2 (Limits) in **Calculus** by Strang & Herman. See openstax.org/books/calculus-volume-1/pages/2-introduction. Section 2.4 (Continuity) relates to today's material.

Continuity

Many functions have the property that you can trace their graphs with pen and paper, without lifting the pen from the page. Such functions are called **continuous**.

Some other functions have points where you have to lift the pen occasionally. We say they have a **discontinuity** at such points.

Intuitively, a function is continuous at a particular point if there is no **break** (or “**jump**”) in its graph at that point.

More formally, we define continuity in terms of **limits**.

Continuity

Definition

A function f is **continuous at** $x = a$ if

1. $f(a)$ is defined, i.e., a is in the domain of f ,
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a} f(x) = f(a)$.

If $f(x)$ is not continuous at $x = a$ we say it is **discontinuous** at $x = a$.

If f is continuous **at every point** in its domain, we say f **is continuous**.

Many functions are continuous, e.g. all polynomial functions, **most** trigonometric functions (not **tan**), $|x|$, and so on.

Example 1

Determine if $f(x) = \frac{x^2 - 4}{x - 2}$ is continuous at $x = 2$.

Example 2

Determine if $f(x) = \begin{cases} 1 - x & x \leq 0 \\ 2 + x & x > 0 \end{cases}$ is continuous at $x = 0$.

Example 3

Determine if $f(x) = \begin{cases} \frac{\sin(x)}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$ is continuous at $x = 0$.

Example

Consider the function

$$f(x) = \begin{cases} x + 1, & x < 2 \\ bx^2, & x \geq 2 \end{cases}$$

For what value of b is f continuous at $x = 2$?

Continuity

Example

For what values of x is $f(x) = \frac{2x + 1}{2x - 2}$ continuous?

Types of discontinuity

In various examples, we have encountered three types of discontinuity.

1. **Removable discontinuity:** $\lim_{x \rightarrow a} f(x)$ exists but

$$\lim_{x \rightarrow a} f(x) \neq f(a)$$

2. **Jump discontinuity:** $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist (and are finite), but $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$

3. **Infinite discontinuity:** At least one of the one-sided limits does not exist.

Types of discontinuity

Example

Each of the following functions has a discontinuity at $x = 2$.
Classify it.

1. $f(x) = \frac{x^2 - 4}{x - 2}$

2. $g(x) = \frac{x^2}{x - 2}$

3. $h(x) = \begin{cases} x/2 & x < 2 \\ -2 & x = 2 \\ x^2 - 3 & x > 2. \end{cases}$

4. $j(x) = \begin{cases} x/2 & x < 2 \\ x^2 - 2 & x > 2. \end{cases}$

Types of discontinuity

Intermediate Value Theorem

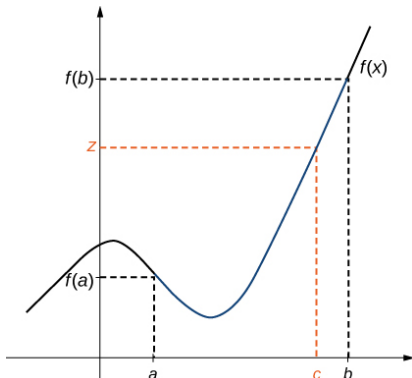
Continuous functions have numerous important properties, many of which we will study in MA140. The first of these is the **Intermediate Value Theorem**.

Intermediate Value Theorem (IVT)

Suppose that $f(x)$ is continuous on an interval $[a, b]$.

Let z be any real number between $f(a)$ and $f(b)$.

Then there exists a number $c \in [a, b]$ such that $f(c) = z$.



- ▶ If you travel by train from Galway to Athlone, then there must be a time when you are at Oranmore station, and a time when you are at Athenry, and at Woodlawn, etc.
- ▶ If your car/train/whatever accelerates from 0km/h to 100km/h , there was a time when it was travelling at 30 km/h .
- ▶ This morning my train ticket from Athenry to Galway cost €5.10. But in January this will be increased to €5.90. But there won't day when it would cost, say, €5.50, because the price had a jump discontinuity (so the IVT does not apply here).

Example

Sketch an example of a function for which the IVT does *not* hold.

One of the main applications of the IVT is in establishing if an equation has a solution:

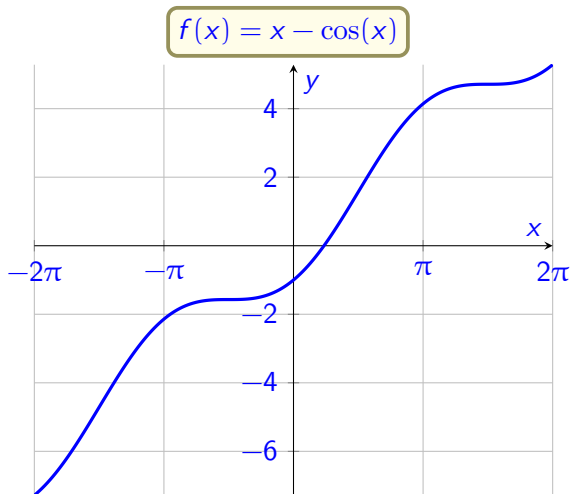
Solutions to $f(x) = 0$

If $f(x)$ defined on $[a, b]$ is such that $f(a) < 0$ and $f(b) > 0$, then there must be a value $c \in [a, b]$ such that $f(x) = 0$.

More generally, if $f(a)f(b) \leq 0$, then $f(x)$ has at least one zero in $[a, b]$.

Example

Show that $f(x) = x - \cos(x)$ has at least one zero.



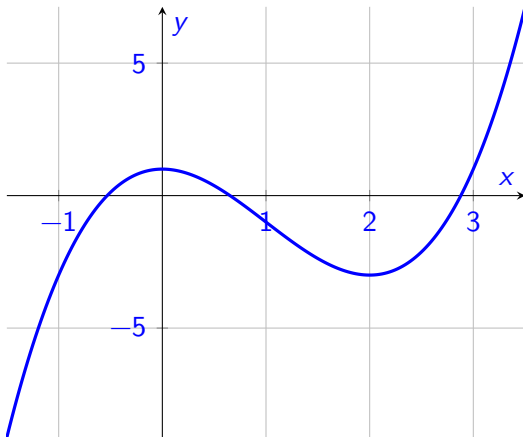
Given a function $f(x)$,

- ▶ When we say c is a **zero** of a function, f , we mean that $f(c) = 0$.
- ▶ Many books and website also use the terminology “ c is a **root** of f ”. This is particularly the case where $f(x)$ is a polynomial.
- ▶ If c is a zero of $f(x)$, then it is a solution to the equation $f(x) = 0$.

Example

How many solutions does $x^3 + 1 = 3x^2$ have?

$$f(x) = x^3 - 3x^2 + 1$$



Example

Use the *Intermediate Value Theorem* to show that the equation

$$2x^3 + 3x^2 - 2x - 1 = 0$$

has three solutions in the range $-2 < x < 1$.

Derivatives: the concept

The next section of MA140 is all about **derivatives** of function. The derivative of a function describes how quickly the function is changing.

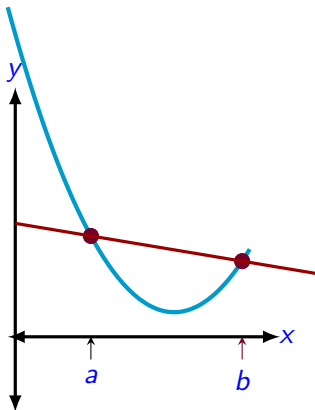
There are many, many applications: derivatives, and equations involving them are used everywhere: **speed/velocity** is the rate of change of displacement; **acceleration** is the rate of change of velocity.

We use derivatives to model how quickly a tumour is growing or shrinking, how pollutants are dispersed in a river, how pressure changes with depth, how inflation is changing in an economy. The list of applications is practically limitless.

Consider the graph opposite. It shows a function, f , and a secant line that intersects f at point $x = a$ and $x = b = a + 2$.

If we wanted to summarised how f is changing between those two values, we could compute it as

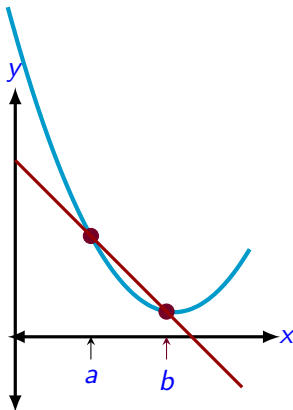
$$\frac{f(b) - f(a)}{b - a} = \frac{f(a + 2) - f(a)}{2}$$



Now we'll consider how f is changing over a shorter interval: from a to $b = a + 1$. Again, we sketch the secant line that intersects f at $x = a$ and $x = b$. The rate of change of f between these two values is

$$\frac{f(b) - f(a)}{b - a} = \frac{f(a + 1) - f(a)}{1},$$

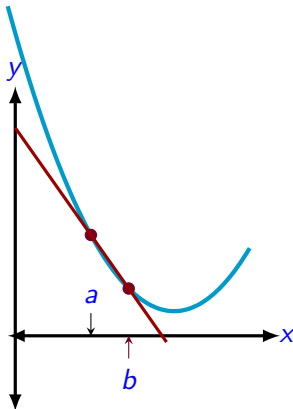
which, of course, is the slope of the secant line.



Next we shorten interval again:
looking at how f changes from
 a to $b = a + \frac{1}{2}$, along with the
secant line that intersects f at
 $x = a$ and $x = b$.

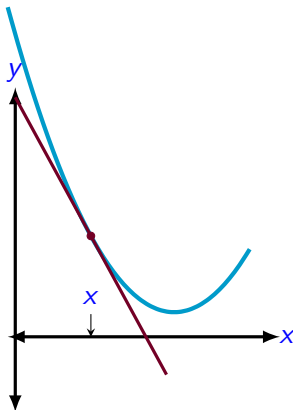
The rate of change of f between
these two values is

$$\frac{f(b) - f(a)}{b - a} = \frac{f(a + \frac{1}{2}) - f(a)}{\frac{1}{2}}.$$



Finally, suppose we want to looking at the **instantaneous** rate of change of f at $x = a$. Hopefully, the preceding images have convinced you we could do this in two (equivalent) ways:

1. $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$
2. or as the slope of the tangent to f at $x = a$.



Exercises 3.3.1 (Based on Q1(a), 23/24)

$$\text{Let } g(x) = \begin{cases} 3 & x \leq 0 \\ 2x + 1 & 0 < x < 1 \\ x^2 & x \geq 1. \end{cases}$$

- (i) Sketch the graph of $g(x)$ on the interval $[-3, 4]$, making use of the empty and full circle notation.
- (ii) Compute $\lim_{x \rightarrow 1^-} g(x)$ and $\lim_{x \rightarrow 1^+} g(x)$. Is g continuous at $x = 1$. If not, classify the type of discontinuity.

Exercises

Exercise 3.3.2

For what values of b and c is $f(x) = \begin{cases} x^2 + 1 & x \leq -1 \\ x + b & -1 < x < 1 \\ cx^2 & x \geq 1. \end{cases}$

continuous at $x = -1$ and $x = 1$?

Exercise 3.3.3 (23/24 exam)

Use the IVT to show that the equation $x^3 - 3x + 1 = 0$ has three solutions in the range $-2 < x < 2$.